Vallabh Vidyasagar

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Education

Pune Institute of Computer Technology Sep 2023 – June 2027
B.E. in Computer Engineering *CGPA: 9.71 / 10*

Experience

Software Development Intern Apr 2026 – Aug 2026
Enable Engineering *Remote*

- Developed a **Flutter based retirement planning application** from scratch across **6 screens**, including login, registration, data entry, analysis, and cashflow visualization.
- Implemented a **computer vision pipeline** using **MediaPipe Pose Landmarker** and **Selfie Segmenter** to estimate user body measurements from a standing image and height input.
- Conducted a comparative study of **17 virtual try-on solutions**, benchmarking **10 commercial APIs** and **3 local models** while evaluating **4 production mobile apps** for feature comparison and market research.
- Translated Figma designs into a production-ready **Flutter** UI redesign spanning **11 screens** across onboarding, avatar creation, shopping, and profile workflows.

Projects

Zenza : Distributed Code Benchmarking Platform | Live | GitHub

- Architected an asynchronous distributed benchmarking platform using **FastAPI** and **Redis**, successfully processing bursts of **500+ concurrent code execution jobs** via scalable worker queues.
- Engineered a secure execution engine using **Docker / Podman**, enforcing strict **CPU and memory resource isolation** to safely compile and run untrusted user code.
- Reduced perceived user latency to **<50ms** by implementing realtime **NDJSON streaming**, allowing the frontend to instantly render status updates as code executes.

micrograd-cpp : C++ Autograd Engine | GitHub

- Reimplemented Andrej Karpathy's **micrograd** in **C++**, building a dynamic computation graph that tracks operations for **automatic differentiation and backpropagation**.
- Designed a lightweight **neural network API** (layers, MLP, loss functions) on top of the autograd engine to support end-to-end model training.
- Managed node lifetimes across the computation graph using `std::shared_ptr`, and validated the engine with example training scripts.

CubeTimer : Rubik's Cube Solves Manager | Flathub | GitHub

- Developed a **GNOME** desktop application featuring **WCA compliant** timing, scramble generation, inspection, and statistical analysis, with a **SQLite** based persistence layer.
- Published the application on **Flathub**, where it currently has **2,900+ active users**.

Skills

Programming Languages: C, C++, Python, Java, Dart
Tools: Docker, Podman, Docker-Compose
Database: MySQL, MongoDB, SQLite, Redis
Backend: FastAPI, Pydantic, Firebase
Frontend: Flutter (cross platform), GTK4 / LibAdwaita (Linux desktop), HTML, CSS
Development Tools: Git, GitHub, GNOME Builder, VS Code
Competitive Programming and DSA: Codeforces, CodeChef, Leetcode

Achievements

- 4th rank in Urban Vision Hackathon hosted by IISc Bangalore.
- 1st rank in PCSB Xenia Codeverse 2026.
- Peak rating of 1411 (Specialist) on Codeforces with a best contest rank of 1284.
- 3 stars on CodeChef with a max rating of 1667 and best contest rank of 35.